Derivative and Integral Table

$f'(x)$	$f(x)$	$\int f(x)\,dx$
0	a	$ax+C$
nx^{n-1}	$x^n\ (n\neq -1)$	$\dfrac{x^{n+1}}{n+1}+C$
$-x^{-2}$	x^{-1}	$\ln\lvert x\rvert+C$
e^x	e^x	e^x+C
$\dfrac{1}{x}$	$\ln x$	$x\ln x-x+C$
$\cos x$	$\sin x$	$-\cos x+C$
$-\sin x$	$\cos x$	$\sin x+C$
$\sec^2 x$	$\tan x$	$\ln\lvert\sec x\rvert+C$
$-\csc x\cot x$	$\csc x$	$-\ln\lvert\csc x+\cot x\rvert+C$
$\sec x\tan x$	$\sec x$	$\ln\lvert\sec x+\tan x\rvert+C$
$-\csc^2 x$	$\cot x$	$\ln\lvert\sin x\rvert+C$
$2\sin x\cos x$	$\sin^2 x$	$\dfrac{x}{2}-\dfrac{\sin 2x}{4}+C$
$-2\sin x\cos x$	$\cos^2 x$	$\dfrac{x}{2}+\dfrac{\sin 2x}{4}+C$
$2\tan x\sec^2 x$	$\tan^2 x$	$\tan x-x+C$
$\dfrac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$	$x\sin^{-1}x+\sqrt{1-x^2}+C$
$-\dfrac{1}{\sqrt{1-x^2}}$	$\cos^{-1} x$	$x\cos^{-1}x-\sqrt{1-x^2}+C$
$\dfrac{1}{1+x^2}$	$\tan^{-1} x$	$x\tan^{-1}x-\dfrac{1}{2}\ln(1+x^2)+C$
$a^x\ln a$	a^x	$\dfrac{a^x}{\ln a}+C$
$x^x(\ln x+1)$	x^x	no elementary form
$\dfrac{1}{x\ln a}$	$\log_a x$	$x\log_a x-\dfrac{x}{\ln a}+C$
$-\dfrac{1}{(x+a)^2}$	$\dfrac{1}{x+a}$	$\ln\lvert x+a\rvert+C$
$-\dfrac{2x}{(x^2+a)^2}$	$\dfrac{1}{x^2+a}$	$\dfrac{1}{\sqrt{a}}\tan^{-1}\left(\dfrac{x}{\sqrt{a}}\right)+C$

Derivative Rules

1. Sum Rule: $[f(x)+g(x)]' = f'(x)+g'(x)$
2. Product Rule: $[f(x)g(x)]' = f'(x)g(x)+g'(x)f(x)$
3. Quotient Rule: $\left[\dfrac{f(x)}{g(x)}\right]' = \dfrac{g(x)f'(x)-f(x)g'(x)}{(g(x))^2}$
4. Chain Rule: $[f(g(x))]' = f'(g(x))\cdot g'(x)$
5. Vector Valued Function: $\mathbf{u}'(t) = (x'(t), y'(t), z'(t))$
6. Dot Product: $[\mathbf{u}\cdot\mathbf{v}]' = \mathbf{u}'\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{v}'$
7. Cross Product: $[\mathbf{u}\times\mathbf{v}]' = \mathbf{u}'\times\mathbf{v}+\mathbf{u}\times\mathbf{v}'$
8. Chain Rule for Partials: $\dfrac{df(x,y)}{dt} = \dfrac{\partial f(x,y)}{\partial x}\dfrac{dx}{dt}+\dfrac{\partial f(x,y)}{\partial y}\dfrac{dy}{dt}$

Jacobian Matrix

$\mathbf{F}:\mathbb{R}^n\to\mathbb{R}^m$, then:

$$\mathbf{F}(x_1,x_2,\ldots,x_n) = \begin{bmatrix} f_1(x_1,x_2,\ldots,x_n) \\ f_2(x_1,x_2,\ldots,x_n) \\ \vdots \\ f_m(x_1,x_2,\ldots,x_n) \end{bmatrix} \qquad \text{Jacobian} = \mathbf{F}' = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

1. Polar Coordinates: $(x,y) = (r\cos\theta, r\sin\theta)$ $\quad |J| = r$
2. Cylindrical Polars: $(x,y,z) = (r\cos\theta, r\sin\theta, z)$ $\quad |J| = r$
3. Spherical Polars: $(x,y,z) = (r\cos\theta\sin\phi, r\sin\theta\sin\phi, r\cos\phi)$ $\quad |J| = r^2\sin\phi$
   $\theta\in[0,2\pi), \phi\in[0,\pi)$

Paramterisations

1. $y = f(x) \to \mathbf{r}(t) = (t, f(t))$
2. $z = f(x,y) \to \mathbf{r}(u,v) = (u,v,f(u,v))$
3. Line by two points **a** and **b**: $\mathbf{r}(t) = \mathbf{a}+t(\mathbf{b}-\mathbf{a}) : 0\le t\le 1$
4. Ellipse: $(x/a)^2+(y/b)^2 = r^2 \to \mathbf{r}(t) = (ar\cos\theta, br\sin\theta)$

5. Ellipsoid: $(x/a)^2+(y/b)^2+(z/c)^2=r^2$
   $\to \mathbf{r}(\theta,\phi)=(ar\cos\theta\sin\phi, br\sin\theta\sin\phi, cr\cos\phi)$

6. Triangle by 3 points: $\mathbf{r}(u,v)=P_0+u(P_1-P_0)+v(P_2-P_0) \quad 0\le v\le 1$ and $0\le u\le 1-v$.

Integration by Parts and Trig Sub

$$\int uv' = uv - \int vu'dx$$

u	v
Polynomial	Exponential or Trig
Log or Inverse Trig	Polynomial

Integral Involves	Substitute	Domain	Identity
$\sqrt{a^2-x^2}$	$x=a\sin u$	$\frac{-\pi}{2}\le u\le\frac{pi}{2}$	$1-\sin^2 u=\cos^2 u$
$\sqrt{a^2+x^2}$	$x=a\tan u$	$\frac{-\pi}{2}\le u\le\frac{pi}{2}$	$1+\tan^2 u=\sec^2 u$
$\sqrt{x^2-a^2}$	$x=a\sec u$	$0\le u\le\frac{pi}{2}$	$\sec^2 u-1=\tan^2 u$

$\sec x=1/\cos x$ and $\csc x=1/\sin x$ and $\cot x=1/\tan x$

Path Integrals

$$\int_C f\,ds=\int_C f(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,dt$$

1. Find parameterisation of C.
2. Compute $f(x,y,z)$ with $x(t),y(t),z(t)$.
3. Find length of $\mathbf{r}'(t)$.
4. Integrate w.r.t t with the bounds from the parameterisation.

Surface Integrals

$$\int_S f\,dS=\iint_S f(\mathbf{R}(u,v))\,|\mathbf{N}(u,v)|\;du\,dv$$

1. Find parameterisation of the surface $S\to\mathbf{R}(u,v)$
2. Plug the parametrisation into f.

3. Find normal vector $\mathbf{N}(u,v) = \dfrac{\partial \mathbf{R}}{\partial u} \times \dfrac{\partial \mathbf{R}}{\partial v}$ and length of normal vector.
4. Integrate w.r.t u, v with bounds from the parameterisation.

Path Integrals of Vector Fields

If $\mathbf{r}'(t) = (a, b)$, then $\mathbf{N}(t) = (-b, a)$.

Circulation: $\int_C \mathbf{F} \cdot \mathbf{dr} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$. Amount of a vector field along a curve.

Flux ($\mathbb{R}^2$): $\int_C \mathbf{F} \cdot \mathbf{dn} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{N}(t) dt$. Amount of a vector field that crosses a curve.

Flux ($\mathbb{R}^3$): $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_S \mathbf{F} \cdot \mathbf{dS} = \iint_D \mathbf{F} \cdot \mathbf{N} du dv$ (see above)

Circulation, Divergence, Curl, Gradient

Circulation is a line integral in a vector field $\int_C \mathbf{F} \cdot \mathbf{dr}$. Divergence is $\nabla \cdot \mathbf{F}$. Curl is $\nabla \times \mathbf{F}$. Gradient is a vector field from a scalar field $\nabla \mathbf{F}$. Flux is the amount of a field that passes through a curve or region.

$\nabla = \left(\dfrac{\partial}{\partial x}, \dfrac{\partial}{\partial y}, \dfrac{\partial}{\partial z} \right)$

Green's Theorem

Relates the circulation around a curve to the double integral of the area inside.

$$\mathbf{F} = (M, N) \quad \oint_c \mathbf{F} \cdot \mathbf{dr} = \iint_D \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} dA \quad (dA = dx dy)$$

Stoke's Theorem

(3d version of Green's) Relates the circulation around a path in 3d to the flux through the enclosed area.

$$\oint_C \mathbf{F} \cdot \mathbf{dr} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{dS}$$

which is a surface integral (see above)

Divergence (Gauss) Theorem

Gauss' theorem relates the outward flux through a surface to a triple integral.

$$\iint_S \mathbf{F} \cdot \mathbf{dS} = \iiint_E \nabla \cdot \mathbf{F} dV \quad (dV = dx dy dz)$$